Harsh Chinmalli

Software Engineer — LLM Systems, RAG & Agentic Workflows

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SUMMARY

AI Engineer with hands-on experience building and shipping production-grade LLM systems, RAG pipelines, and agentic workflows. Proficient in prompt engineering, context engineering, retrieval-augmented generation, NLP, and inference optimization. Built AI-powered automation agents, multi-step agentic systems with tool orchestration, and LLM-grounded applications on GCP, AWS, and Azure. Strong first-principles thinker who reasons about why AI systems fail — not just how to prompt them. Experienced with evaluation harnesses, RLHF feedback loops, and deploying reliable AI in compliance-aware enterprise environments.

EDUCATION

KLE Technological University 2022 – 2025

Bachelor of Computer Applications, Hubli

EXPERIENCE

Cognizant Technology Solutions Sep 2025 – Present

AI Augmented Infra Ops Developer *Bangalore, India*

- **Designed and shipped agentic AI workflows** to autonomously resolve L1.5–L2 IT support tickets — multi-step systems that use tools, retrieve context, and take decisions with minimal human intervention.
- **Built a Global Cloud Health Dashboard** using LLM-powered inference and real-time data retrieval to monitor cloud provider status and surface actionable operational insights.
- **Developed a Major Incident Management (MIM) agentic bridge** — an AI-orchestrated system that detects P1 incidents, triggers meeting bridges, notifies stakeholders, and logs RCA context automatically.
- **Engineered a self-service onboarding agent** with context-aware NLP intake, PostgreSQL-backed state management, and automated server registration into the enterprise automation platform.
- **Applied prompt engineering and context engineering** techniques across enterprise AI features to improve LLM output reliability, reduce hallucinations, and align model behaviour with business guardrails.

Sikoter / XcelUp (LLC) Jan 2025 – Sep 2025

Head of AI, Product & Learning (Remote) *Chicago, IL*

- **Built and shipped XEVA AI** — a production RAG system using Azure AI Search and Azure Cognitive Search to ground LLM responses in platform-specific knowledge, enabling intelligent user guidance and context-aware answers.
- **Designed end-to-end RAG pipelines** with document ingestion, chunking strategies, embedding generation, vector retrieval, and response synthesis — optimizing for retrieval quality, latency, and cost.
- **Built AI-powered agentic workflows** for content generation, multilingual translation, community moderation, and customer onboarding — multi-step LLM orchestration with tool use and RLHF-informed feedback loops.
- **Implemented LLM observability and AIops practices** using Winston.js and centralized logging to monitor inference quality, track model outputs, and enable proactive debugging of AI system failures.
- **Contributed to SOC 2 and payments-grade compliance** by implementing guardrails, security-focused application changes, and audit-ready controls for enterprise customer onboarding.
- **Automated CI/CD pipelines** via GitHub Actions and optimized database queries on GCP, improving deployment reliability and infrastructure performance by 30%.

SKILLS

AI Core: LLMs, RAG (Retrieval-Augmented Generation), Agentic AI, Prompt Engineering, Context Engineering, NLP, Inference Optimization, RLHF / Feedback Loops, Guardrails

AI Tooling & Ops: LangChain, FAISS, Azure AI Search, Azure Cognitive Search, Gemini API, Evaluation Harness Engineering, AIops, MCP (Model Context Protocol)

Languages: Python, JavaScript, TypeScript, Java, Go, C/C++

Backend & APIs: Node.js, Express.js, REST APIs, Microservices, OpenAPI/Swagger

Databases & Retrieval: PostgreSQL, MongoDB, MySQL, DynamoDB, Vector Stores

Cloud & DevOps: GCP, AWS, Azure, Docker, Kubernetes, CI/CD (GitHub Actions), Linux

Frontend: React.js, Next.js, Tailwind CSS, HTML, CSS

Practices: Agile/Scrum, On-Call Operations, Incident Management, Git, Postman

PROJECTS

MCP Gateway | Python, FAISS, Groq/Gemini APIs, SQLite | GitHub

- **Built a protocol-level Python proxy** between AI agents and 150+ tools across real and simulated MCP (Model Context Protocol) servers, using FAISS semantic retrieval to cut prompt tokens 27× (19,167 → 701) and end-to-end latency 26% at scale, while holding tool-selection accuracy at 97.9%.
- **Ran a controlled benchmark** (48 labelled test cases, 300+ live API calls) across three setups and a 10–150 tool crossover sweep, empirically identifying the tool-count threshold above which retrieval pays for itself.
- **Implemented a policy layer** with SHA-256 description hashing to detect silent tool tampering, prompt-injection scanning, and a SQLite audit log of every call – verified live against a security demo.
- **Packaged the project** as an installable Python library (pyproject.toml), consumed as a dependency in a second, downstream project.

PDF Intelligence Engine — Production RAG Pipeline

- **Built an offline RAG system** that answers natural language questions over PDF documents using a full retrieval pipeline: document ingestion, semantic chunking, embedding generation, and context-grounded response synthesis.
- **Integrated LangChain** for orchestration of the document chunking, embedding, and LLM query pipeline — with context engineering techniques to maximise answer relevance and minimise hallucination.
- **Used FAISS vector store** for semantic similarity search and accurate context passage retrieval, with inference optimized for low-latency local execution and zero external server dependency.
- **Designed the system with evaluation harness principles** — iterating on chunking strategy, retrieval depth, and prompt templates to improve answer faithfulness and groundedness on domain-specific documents.

XEVA AI — Agentic LLM Platform Assistant

- **Designed and shipped an agentic AI assistant** using Azure AI Search and Cognitive Search as the retrieval backbone, enabling the LLM to ground answers in platform-specific knowledge and guide users through complex workflows.
- **Built multi-step agentic workflows** for content generation, translation, and onboarding — LLM orchestration with tool use, context injection, and RLHF-informed output refinement.
- **Implemented prompt engineering and context engineering** strategies to control LLM behaviour, enforce compliance guardrails, and align outputs with business-specific constraints in a payments-adjacent environment.

CodeClub — LLM-Powered Assessment Engine

- **Integrated Gemini API** with prompt engineering and NLP techniques to automatically generate coding questions, test cases, and evaluation rubrics for instructors — grounding outputs in programming domain knowledge.
- **Built an evaluation harness** to benchmark LLM-generated question quality, iterating on prompt templates and context injection strategies to improve output consistency and difficulty calibration.
- **Engineered a scalable platform** with leaderboards and performance tracking, designed for gamification and future AI-powered personalised learning paths.

BlogTech — Secure Internal Q&A Platform

- Built a secure internal Q&A platform using Node.js, Express, and MongoDB to streamline employee knowledge sharing.
- Integrated role-based access and JWT auth to ensure secure multi-level user interactions.
- Implemented real-time error logging with Winston + MySQL, enabling proactive debugging and system monitoring.

CERTIFICATIONS

Context Engineering Foundation — Cognizant
Introduction to Model Context Protocol — Anthropic
Introduction to Subagents — Anthropic
Building with Claude API — Anthropic
Build with Gemini — Google Cloud





LANGUAGES KNOWN

English, Hindi, Kannada